

Task 10: Session 10 (AIML) - Assignment Questions

Name: Mihir Sopan Patil

Domain: AIML

College & Branch Name: Zeal Polytechnic, Diploma In AIML

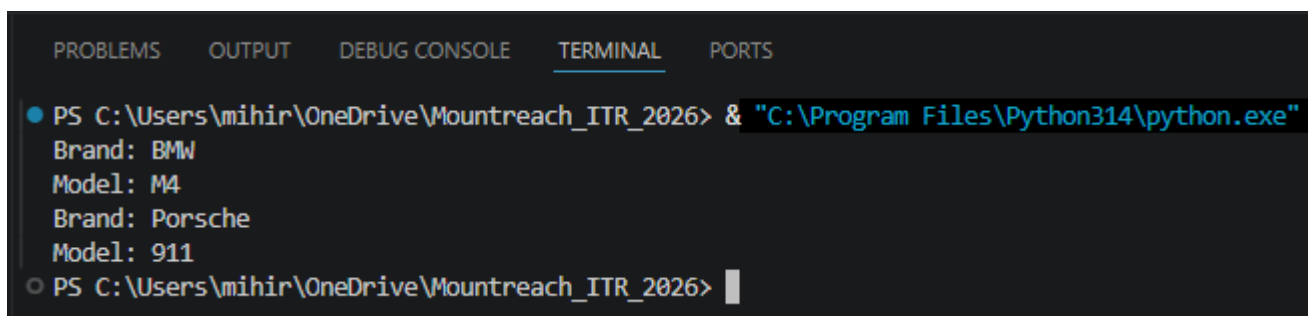
1. Basic Class & Object

Create a class named Car with:

- Attributes: brand and model (set using `__init__`).
- A method `display_info()` that prints the brand and model.

Create two objects of the Car class with different values and call the method for both.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Brand: BMW
Model: M4
Brand: Porsche
Model: 911
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

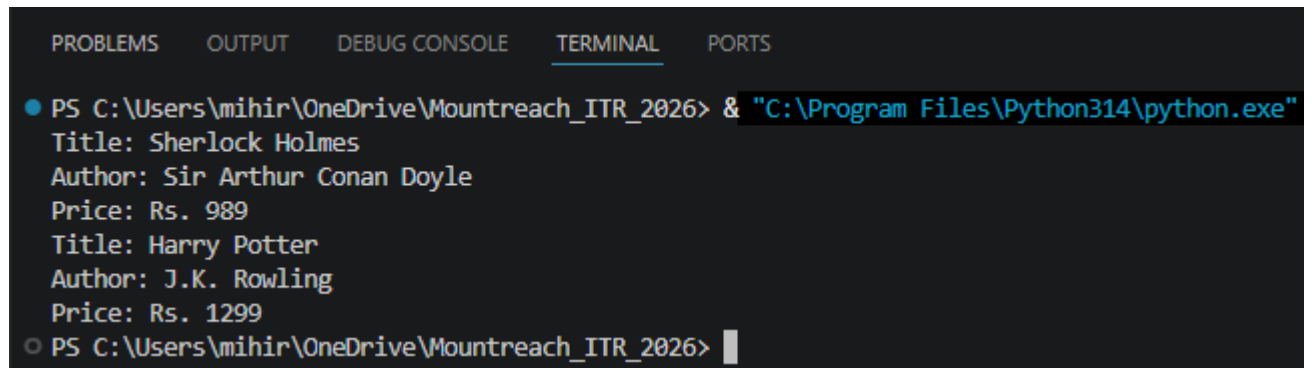
2. Using init Constructor

Write a class Book that takes title, author, and price as parameters in `__init__`.

Create a method `show_details()` to print all the book information.

Create at least two book objects and display their details.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Title: Sherlock Holmes
Author: Sir Arthur Conan Doyle
Price: Rs. 989
Title: Harry Potter
Author: J.K. Rowling
Price: Rs. 1299
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

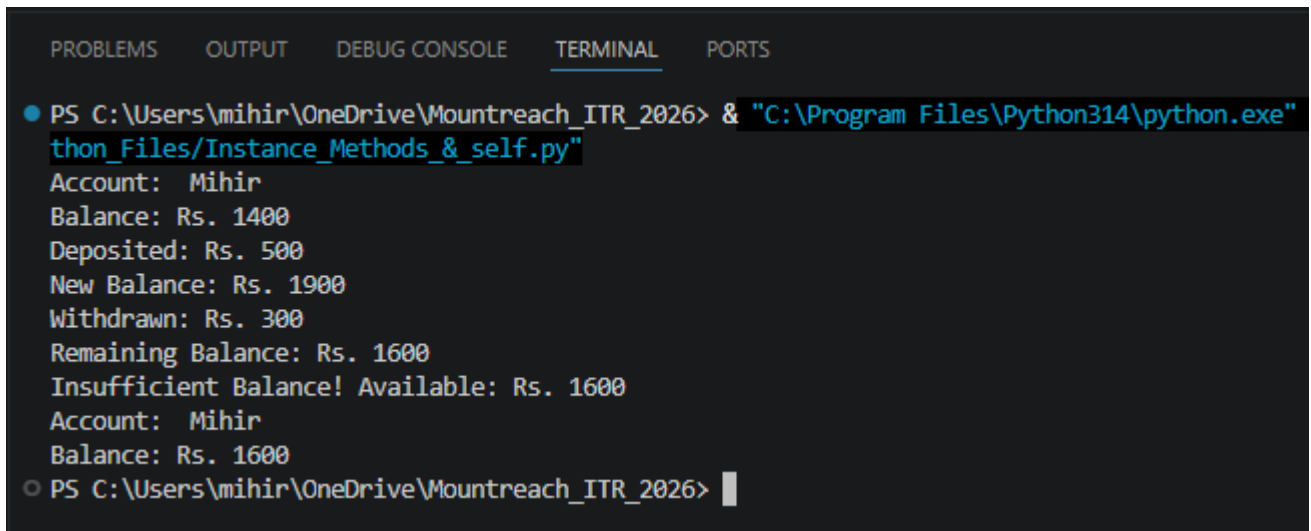
3. Instance Methods & self

Create a class BankAccount with:

- Attributes: `account_holder` and `balance` (initialized in `__init__`).
- Methods:
 - `deposit(amount)` — add amount to balance.
 - `withdraw(amount)` — subtract amount from balance (only if sufficient balance).
 - `show_balance()` — print current balance.

Create an object and demonstrate all methods.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  thon_Files/Instance_Methods_&_self.py"
Account: Mihir
Balance: Rs. 1400
Deposited: Rs. 500
New Balance: Rs. 1900
Withdrawn: Rs. 300
Remaining Balance: Rs. 1600
Insufficient Balance! Available: Rs. 1600
Account: Mihir
Balance: Rs. 1600
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

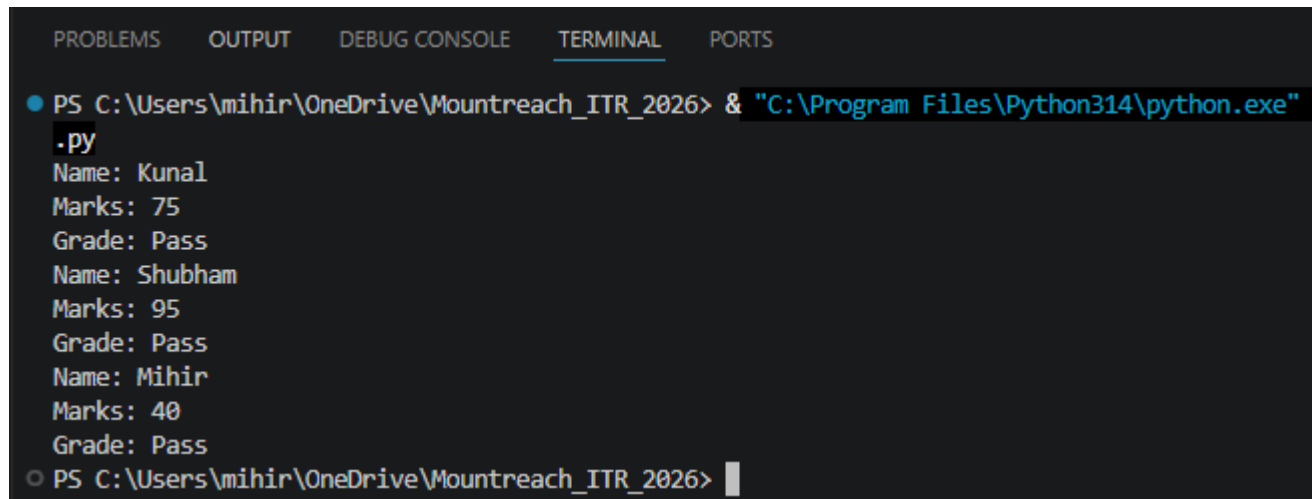
4. Method With Parameters

Create a class Student with attributes name and marks.

Add a method calculate_grade() that returns "Pass" if marks ≥ 40 , otherwise "Fail".

Create 3 student objects, display their details, and print their grades.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  .py
  Name: Kunal
  Marks: 75
  Grade: Pass
  Name: Shubham
  Marks: 95
  Grade: Pass
  Name: Mihir
  Marks: 40
  Grade: Pass
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

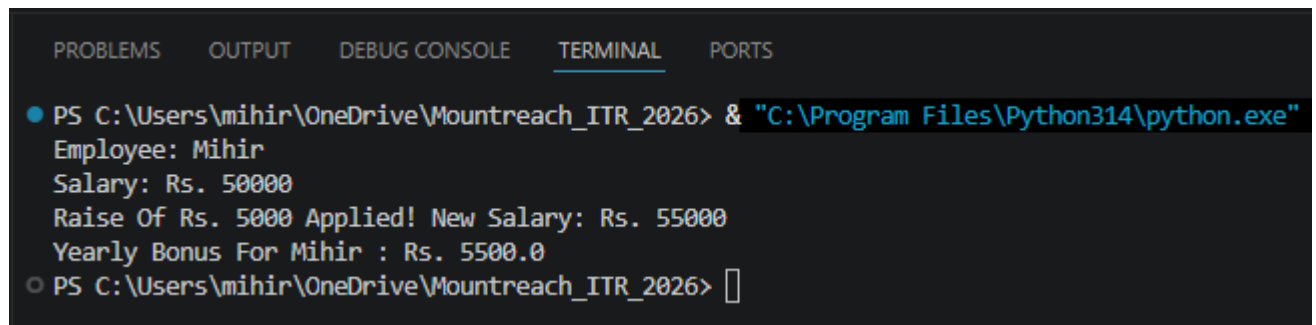
5. Multiple Methods

Create a class Employee with:

- Attributes: name, salary (use `__init__`).
- Methods:
 - `display_details()`
 - `give_raise(amount)` — increase salary and print new salary.
 - `yearly_bonus()` — return 10% of current salary.

Create an employee object and demonstrate all methods.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Employee: Mihir
Salary: Rs. 50000
Raise Of Rs. 5000 Applied! New Salary: Rs. 55000
Yearly Bonus For Mihir : Rs. 5500.0
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> █
```

6. Real - World Object Modeling

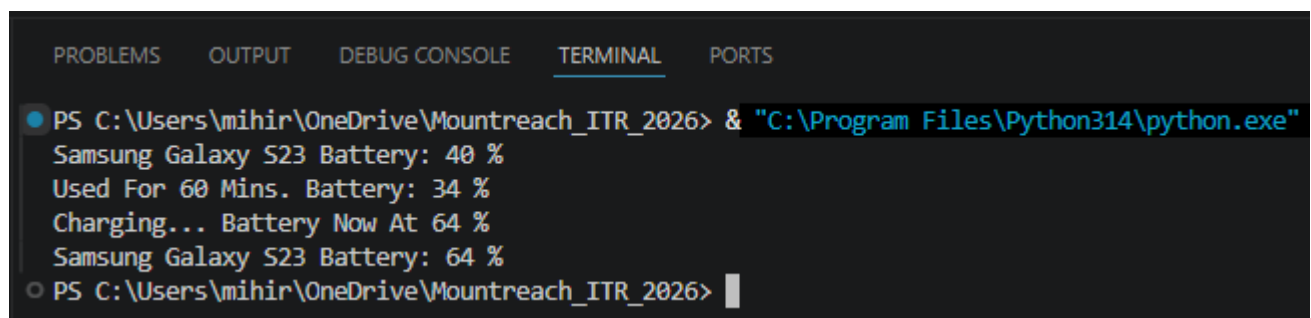
Create a class `MobilePhone` with attributes: `brand`, `model`, `battery_percentage`.

Add these methods:

- `charge(percent)` — increase battery.
- `use_phone(minutes)` — reduce battery by 1% per 10 minutes.
- `show_status()` — print current status.

Create an object and simulate some usage.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Samsung Galaxy S23 Battery: 40 %
Used For 60 Mins. Battery: 34 %
Charging... Battery Now At 64 %
Samsung Galaxy S23 Battery: 64 %
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

7. Combining Concepts

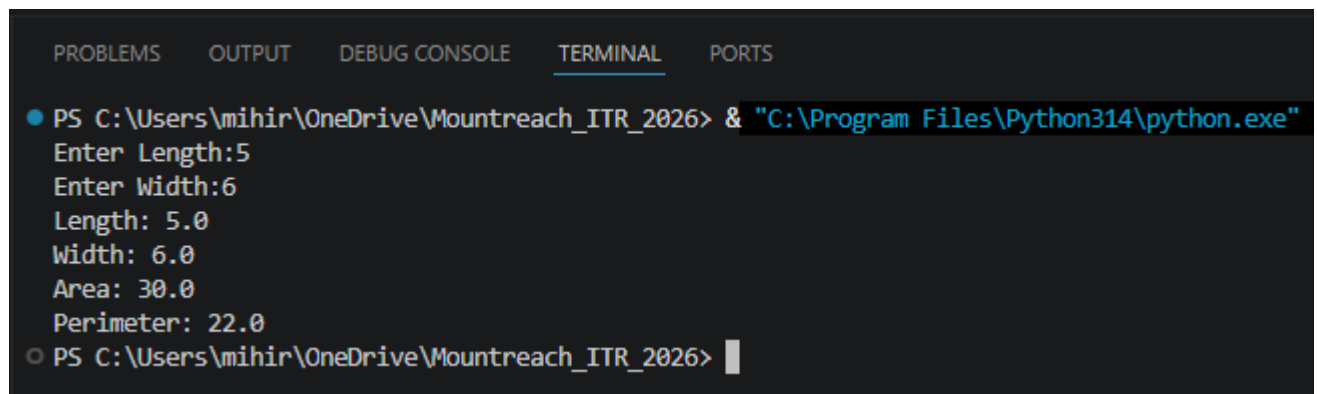
Create a class Rectangle with attributes length and width.

Add methods:

- `area()` — return the area.
- `perimeter()` — return the perimeter.
- `display()` — print length, width, area, and perimeter.

Take input from user for length and width, create object, and display results.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter Length:5
Enter Width:6
Length: 5.0
Width: 6.0
Area: 30.0
Perimeter: 22.0
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

8. Updating Attributes

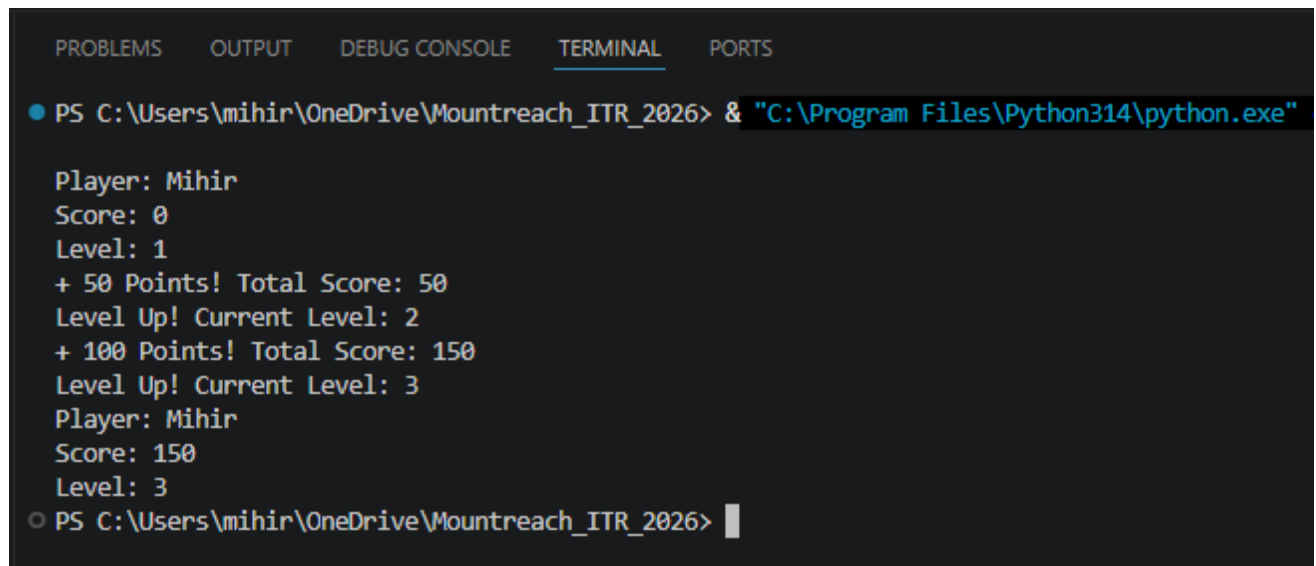
Create a class Player with attributes name, score, and level.

Add methods:

- increase_score(points)
- level up () - increase level by 1.
- show_progress()

Create an object, update score and level multiple times, and show progress.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

Player: Mihir
Score: 0
Level: 1
+ 50 Points! Total Score: 50
Level Up! Current Level: 2
+ 100 Points! Total Score: 150
Level Up! Current Level: 3
Player: Mihir
Score: 150
Level: 3
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```


9. Debugging OOP Code

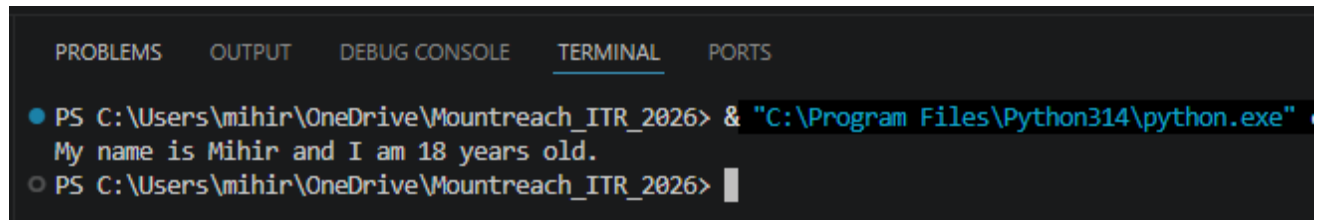
The following code has errors. Correct it and add proper comments explaining the fixes:

```
class Person:
    def __init__(name, age):
        name = name
        age = age

    def introduce():
        print("My name is" name "and I am" age "years old.")

p1 = Person("Rahul", 25)
p1.introduce()
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
My name is Mihir and I am 18 years old.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

10. Mini Project - Combined Concepts

Create a **Simple Library Management System** using OOP:

Design a class Book with:

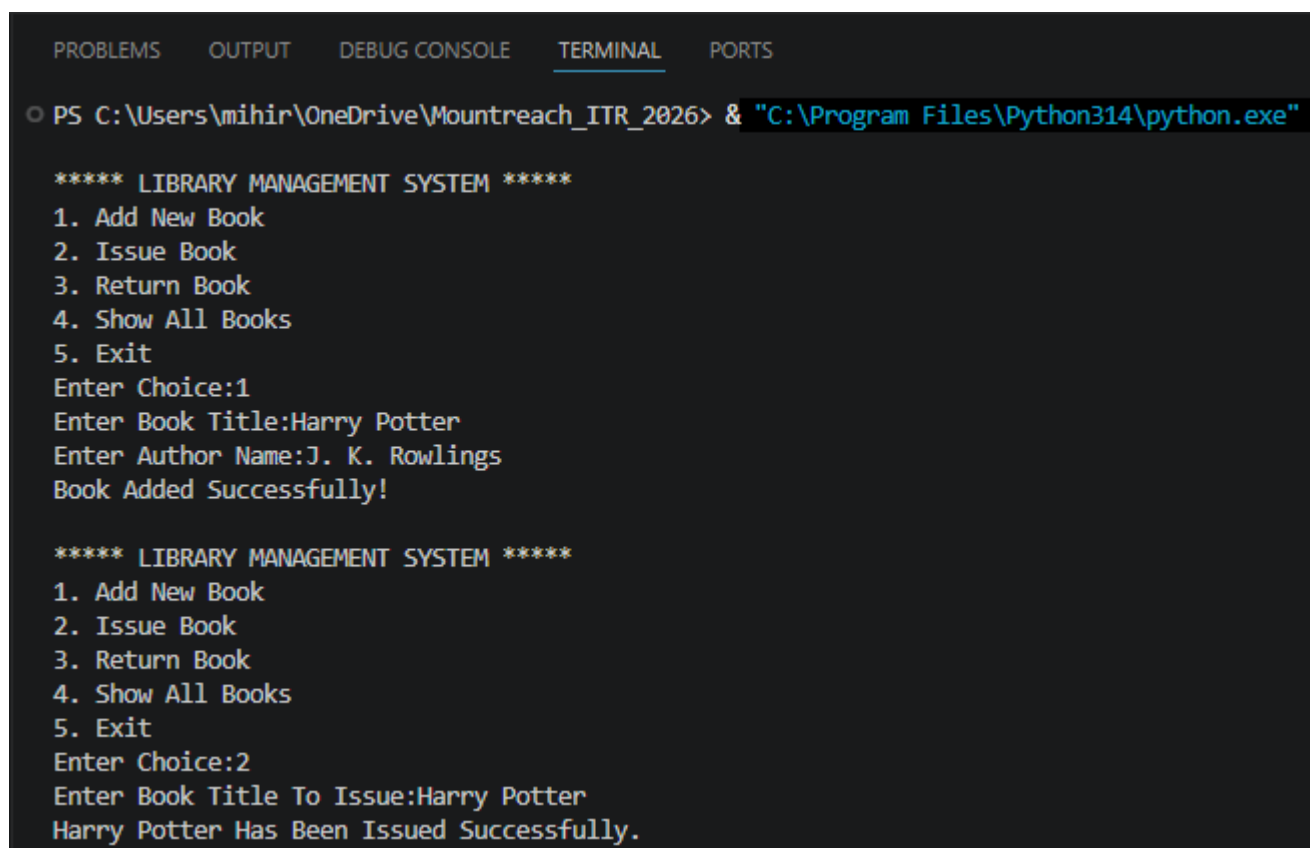
- Attributes: title, author, status ("Available" or "Issued").
- Methods: issue_book(), return_book(), show_info().

Create multiple book objects and demonstrate issuing/returning books with proper messages.

Use a menu (while loop) to allow the user to:

- a) Add a new book
- b) Issue a book
- c) Return a book
- d) Show all books
- e) Exit

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"

***** LIBRARY MANAGEMENT SYSTEM *****
1. Add New Book
2. Issue Book
3. Return Book
4. Show All Books
5. Exit
Enter Choice:1
Enter Book Title:Harry Potter
Enter Author Name:J. K. Rowlings
Book Added Successfully!

***** LIBRARY MANAGEMENT SYSTEM *****
1. Add New Book
2. Issue Book
3. Return Book
4. Show All Books
5. Exit
Enter Choice:2
Enter Book Title To Issue:Harry Potter
Harry Potter Has Been Issued Successfully.
```